

PROJECT REPORT FOR INDUSTRIAL TRAINING AND INTERNSHIP



AUTODESK
AutoCAD

SUBMITTED IN 3D MODEL OF FIGHTER JET PLANE USING AUTOCAD

At



**ARDENT
COMPUTECH
PVT. LTD.**

FROM 16 March 2025 TO 16 MAY 2025



SUBMITTED BY: -

- **SUBRATA PANDA**
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- **SUMAN RAJAK**
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- **PAPUN PATRA**
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PROJECT SUPERVISOR

DEPARTMENT OF MECHANICAL ENGINEERING
RAMKRISHNA MAHATO GOVERNMENT ENGINEERING COLLEGE
PURULIA

3D MODEL OF A FIGHTER JET PLANE USING AUTOCAD

A Project Report for Industrial Training and Internship

Submitted by

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In the partial fulfillment of the award of the degree of

B.Tech

in the
Department

of

MECHANICAL ENGINEERING

**RAMKRISHNA MAHATO GOVERNMENT
ENGINEERING COLLEGE , PURULIA**



AT

Ardent Computech Pvt. Ltd.





CERTIFICATE FROM SUPERVISOR

This is to certify that :

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have completed the project titled " **3D MODEL OF A FIGHTER JET PLANE USING AUTOCAD** " under my supervision during the period from "16/03/2025" to "16/05/2025" which is in partial fulfillment of requirements for the award of the **B.Tech** degree and submitted to the Department of "**MECHANICAL ENGINEERING**" of "**RAMKRISHNA MAHATO GOVERNMENT ENGINEERING COLLEGE , PURULIA.**"

Suman Kalyan Das.

Signature of the Supervisor

Date: 16.05.2025

Name of the Project Supervisor: SUMAN KALYAN DAS





BONAFIDE CERTIFICATE

Certified that this project work was carried out under my supervision

“ 3D MODEL OF A FIGHTER JET PLANE USING AUTOCAD ” is the bonafide work

SL NO	NAME OF THE STUDENT	SIGNATURE
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2.	JYOTIRMOY MAHATA	
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PROJECT MENTOR



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We take this opportunity to express our deep gratitude towards our project mentor, **SUMAN KALYAN DAS** for giving such valuable suggestions, guidance and encouragement during the development of this project work.

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PROJECT OBJECTIVE

The objective of the training in AutoCAD 2D & 3D is to learn how to create basic 2D drawings then gradually complex 3D drawings in the AutoCAD software. The AutoCAD software is one of the most sophisticated computer applications that we are likely to encounter. Therefore, learning to use it can be challenging. Learning AutoCAD to use daily for drawing creation and editing, or simply need to understand how to navigate around a drawing file, this training provided me all.

1. Visualization: To create a detailed and realistic representation of a fighter jet, showcasing its design and features.
2. Design Evaluation: To evaluate the aerodynamic design, shape, and structure of the fighter jet.
3. Artistic Expression: To demonstrate artistic skills and creativity in 3D modeling and rendering.
4. Educational Purpose: To teach or learn about the design, engineering, and functionality of fighter jets.
5. Simulation and Analysis: To prepare the model for simulations, such as aerodynamic analysis or flight simulations.

INTRODUCTION

- A fighter jet model in AutoCAD involves designing the external body, wings, cockpit, and structural framework using various 3D modeling techniques. This process can be broken down into several key steps:
- **Understanding the Design** – Research the aerodynamics, dimensions, and structure of real fighter jets to create an accurate representation.
- **Setting Up AutoCAD** – Choose the correct workspace (3D Modeling) and configure units for precise measurements.
- **Basic 3D Shapes and Modeling** – Start by creating the fuselage using extrusions, lofts, or sweeps.
- **Refining Details** – Add wings, cockpit, tail fins, and landing gear using additional 3D operations like chamfering, filleting, and Boolean operations.
- **Applying Materials and Rendering** – Use AutoCAD's rendering tools to add textures and visualize the jet in a realistic environment.



Advantages of using jet plane :

- Fighter jets are advanced military aircraft designed for speed, agility, and combat effectiveness. Here are some key advantages:
- **Superior Speed and Maneuverability** – Fighter jets can travel at supersonic speeds, making them highly effective in aerial combat and quick response missions.
- **Enhanced Combat Capabilities** – Equipped with advanced weaponry, radar systems, and defense mechanisms for engaging enemy targets with precision.
- **Strategic Military Defense** – Essential for national security, fighter jets help deter threats and provide aerial dominance in warfare.
- **Multirole Functionality** – Modern fighter jets can perform various roles, including air superiority, ground attack, reconnaissance, and electronic warfare.
- **Advanced Technology Integration** – Features like stealth, AI-assisted targeting, and next-gen avionics enhance combat effectiveness

Disadvantages of using jet plane :

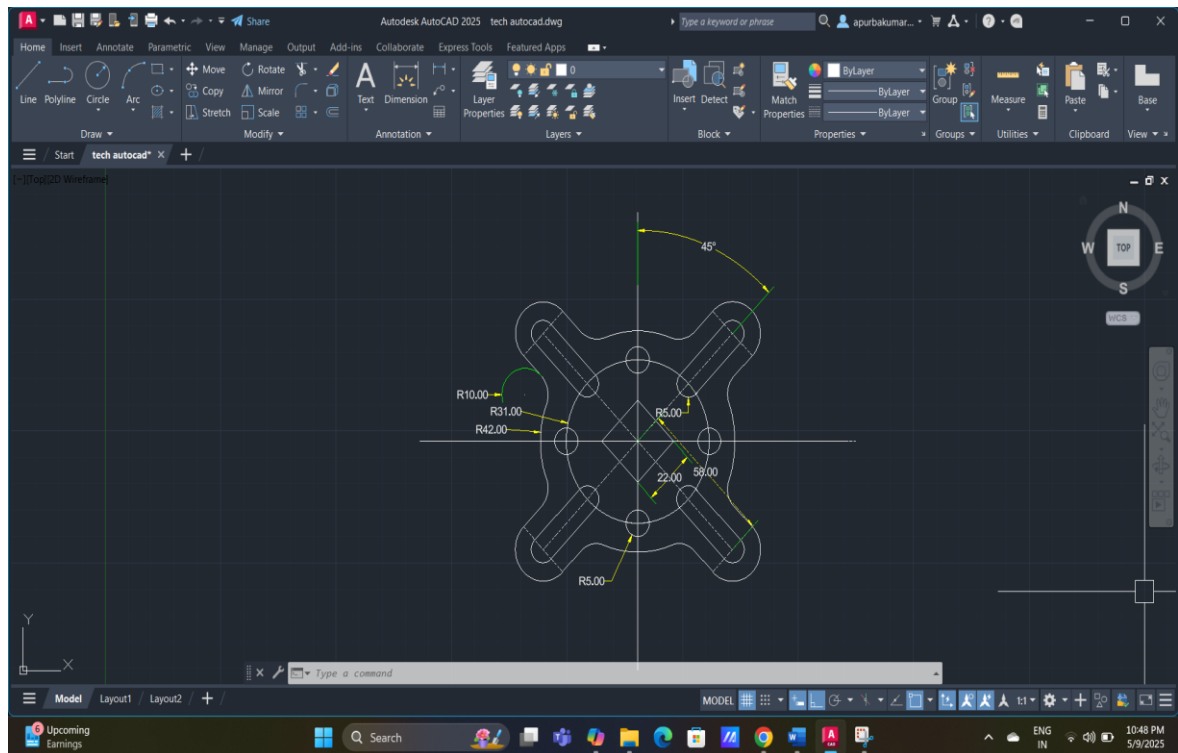
- **High Cost** – Development, production, and maintenance require substantial financial investment.
- **Fuel Consumption** – Fighter jets consume large amounts of fuel, making them expensive to operate.
- **Limited Range** – Without aerial refueling, their operational range is restricted.
- **Environmental Impact** – High fuel consumption contributes to carbon emissions and noise pollution.
- **Pilot Vulnerability** – Despite advanced defensive systems, pilots remain exposed to risks in combat situations.

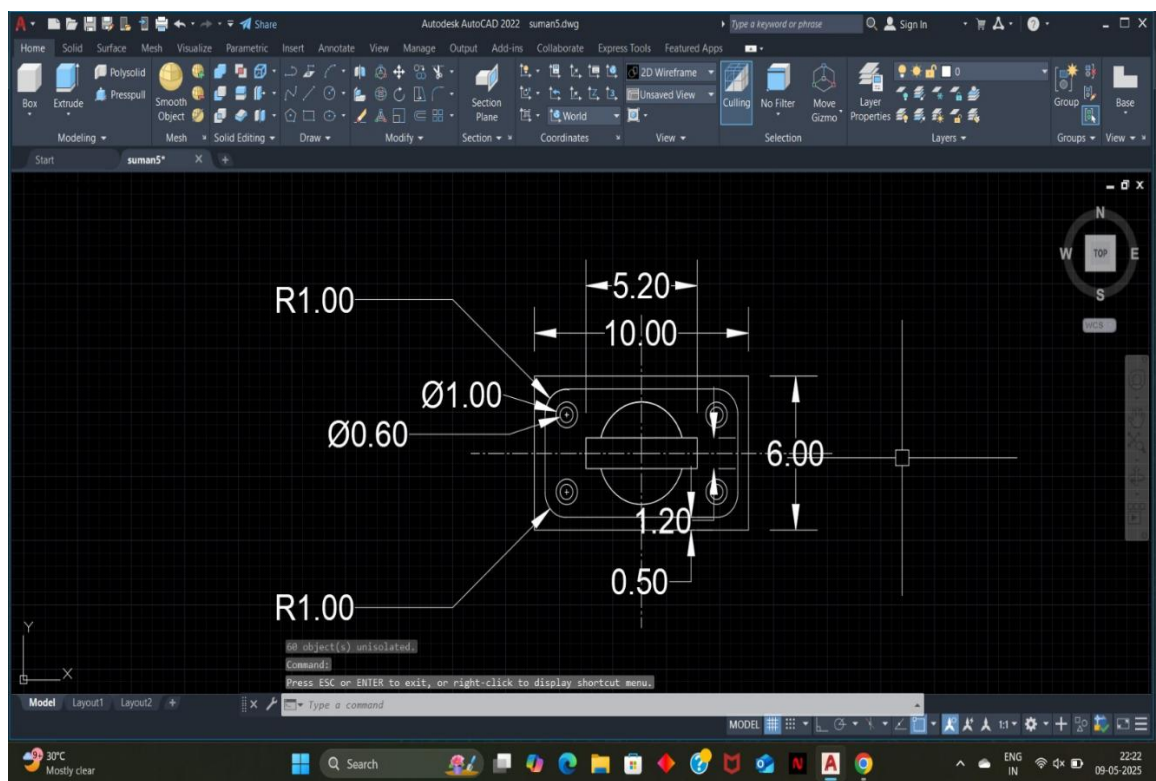
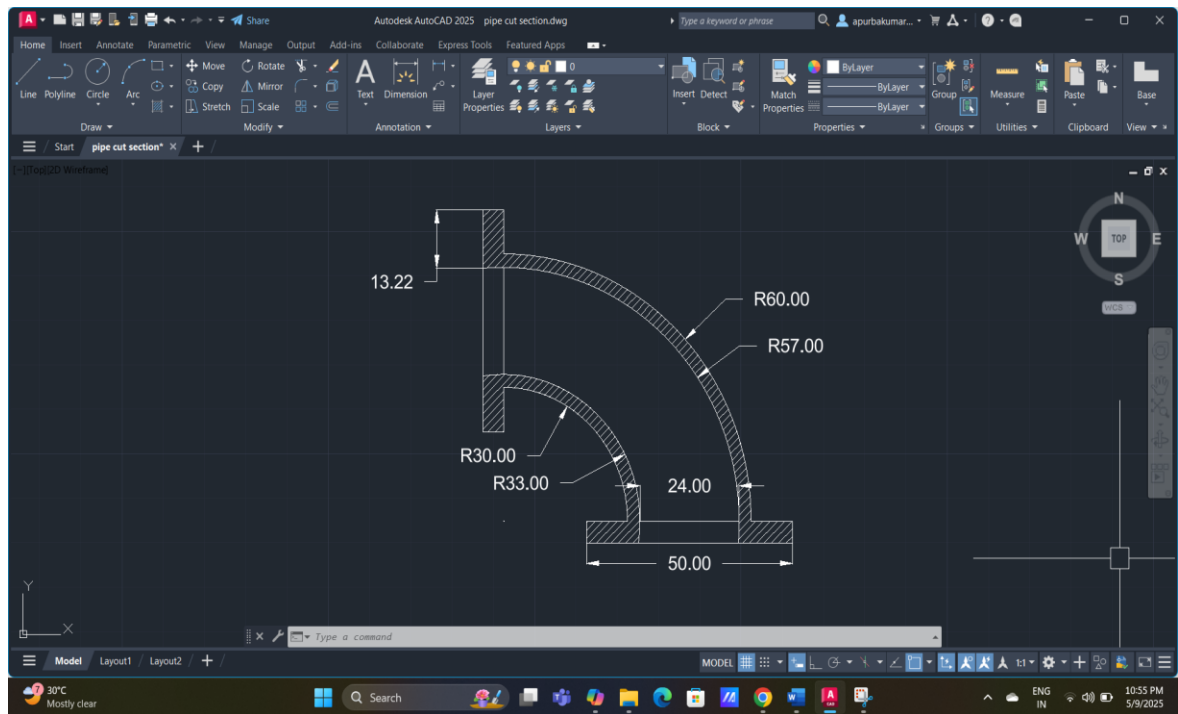
- **Complex Maintenance** – Requires specialized teams for upkeep, making maintenance time-consuming and costly.

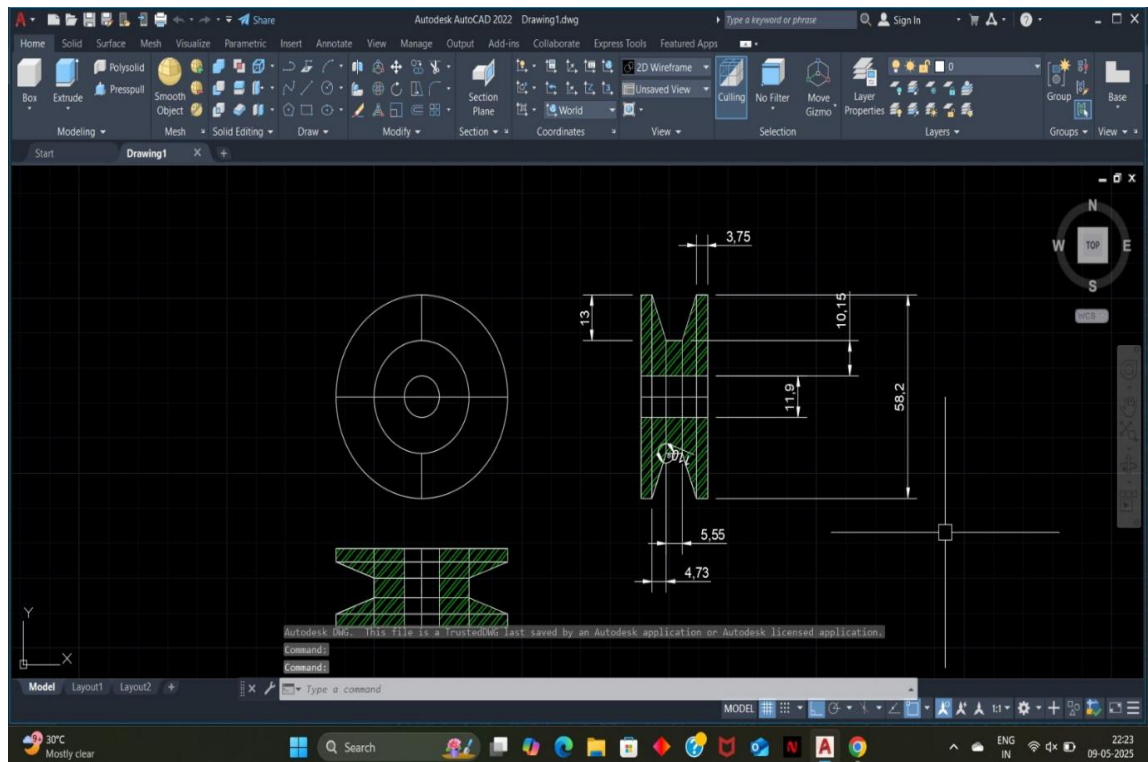
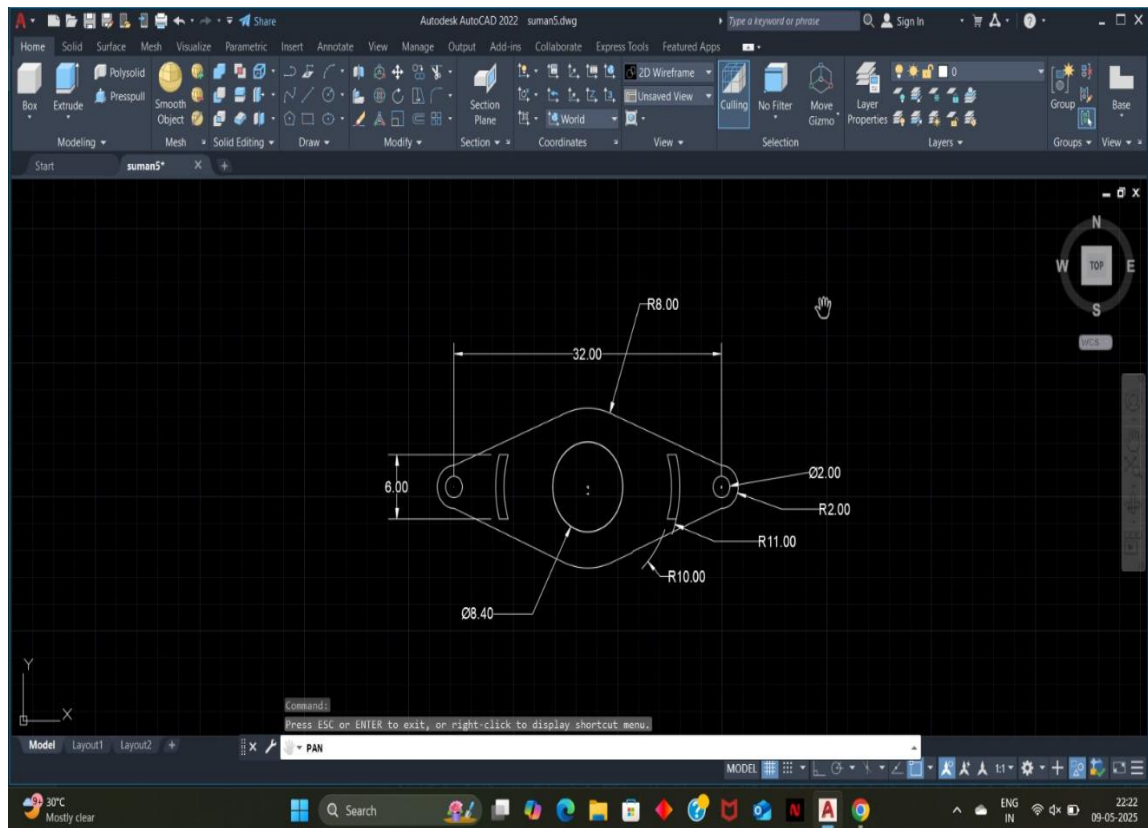
Applications of using jet plane :

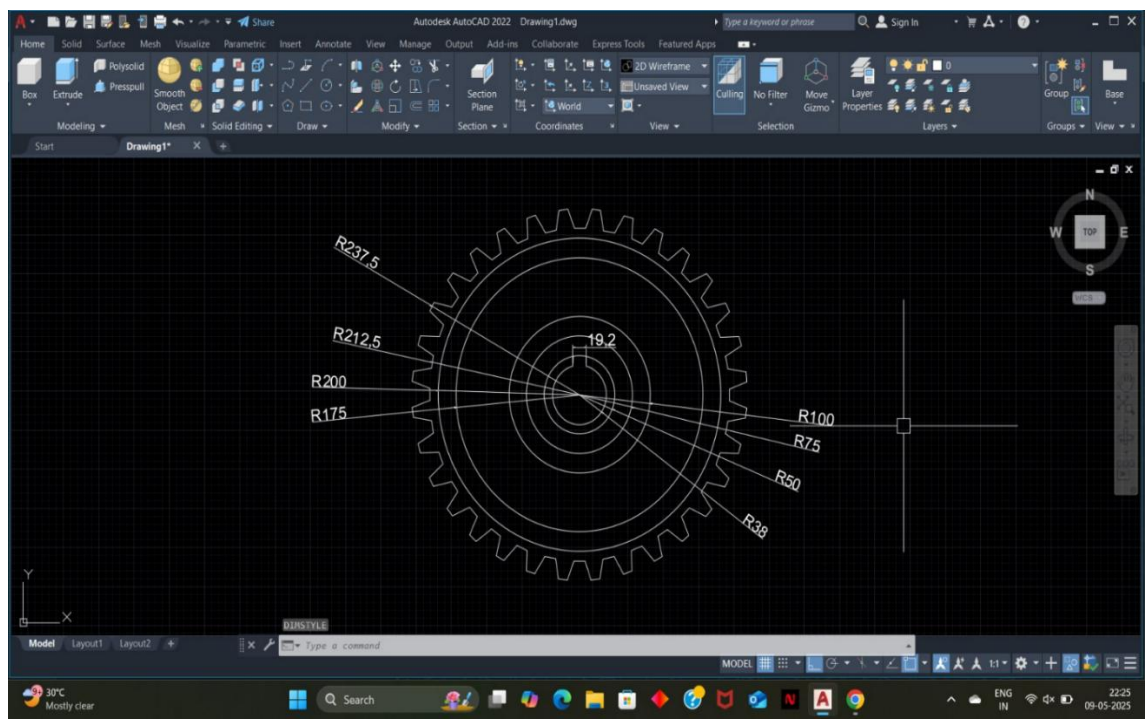
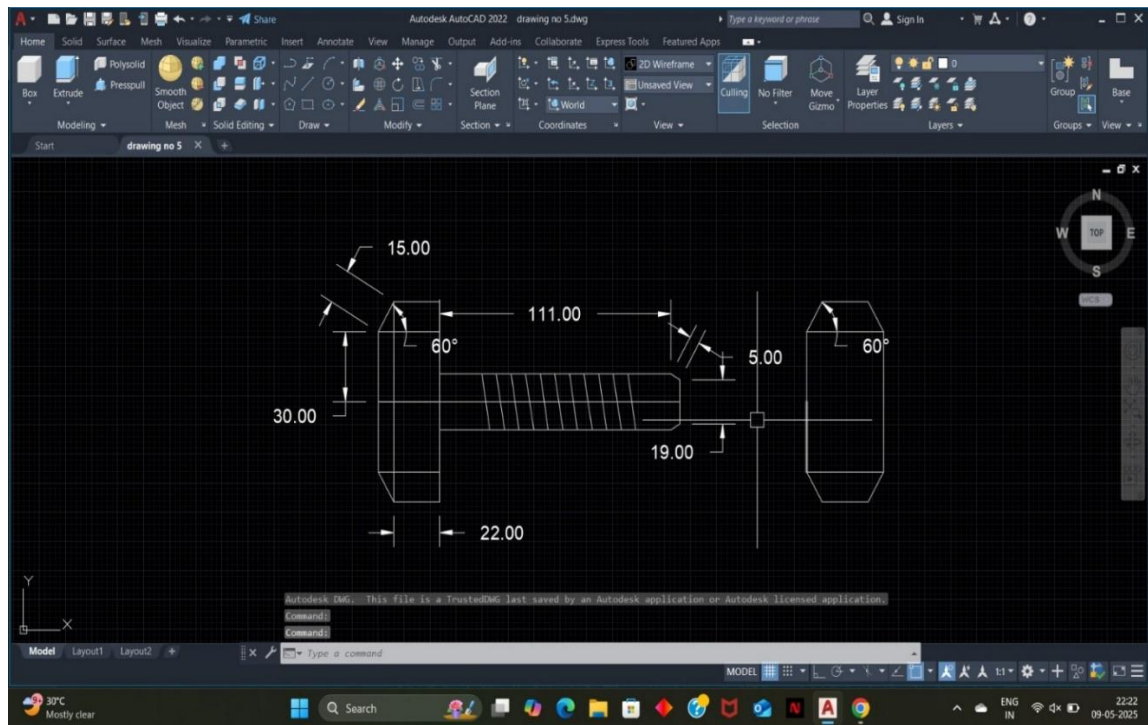
- ❖ **Air Superiority & Defense** – Ensuring control of airspace by intercepting enemy aircraft and conducting defensive operations.
- ❖ **Ground Attack & Support** – Engaging ground targets to support military forces during warfare.
- ❖ **Reconnaissance & Surveillance** – Equipped with advanced sensors to gather intelligence on enemy movements and terrain.
- ❖ **Aerial Patrol & Border Security** – Used for patrolling national borders and detecting unauthorized aircraft.
- ❖ **Electronic Warfare** – Disrupting enemy communications and radar systems using specialized technology.
- ❖ **Training & Research** – Fighter jets help train pilots and assist in aerodynamics and aviation research.

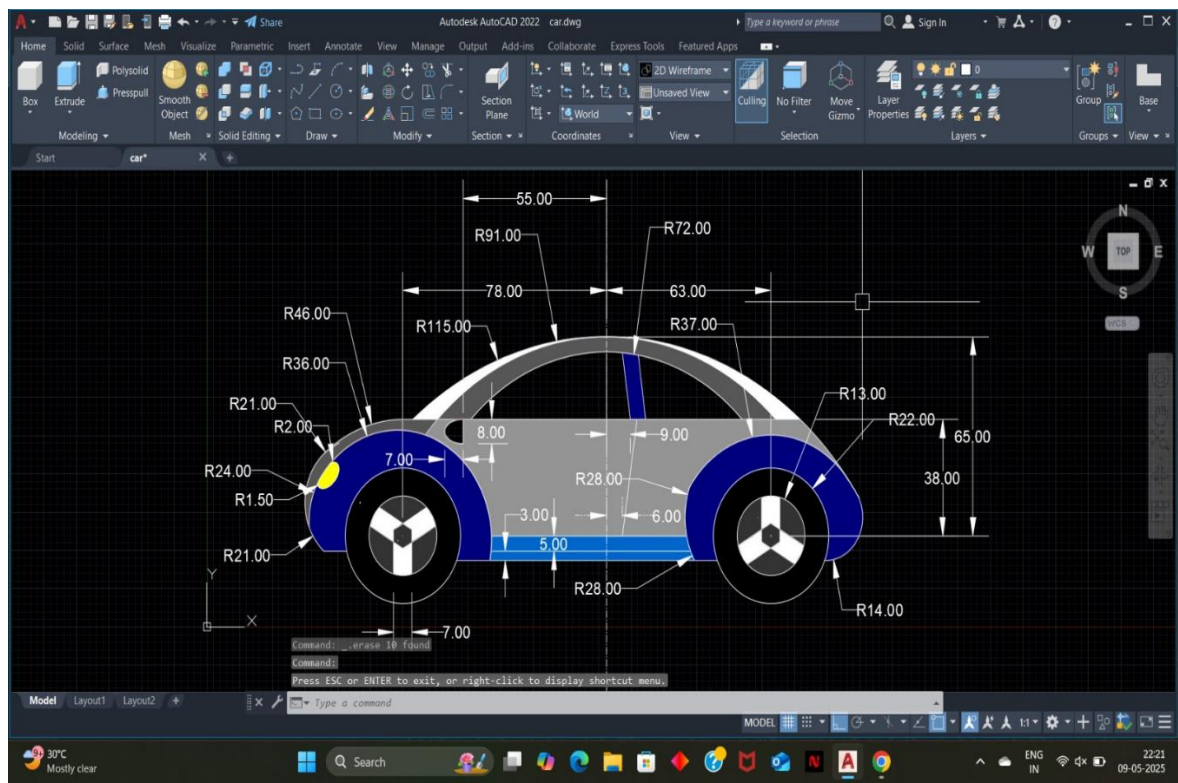
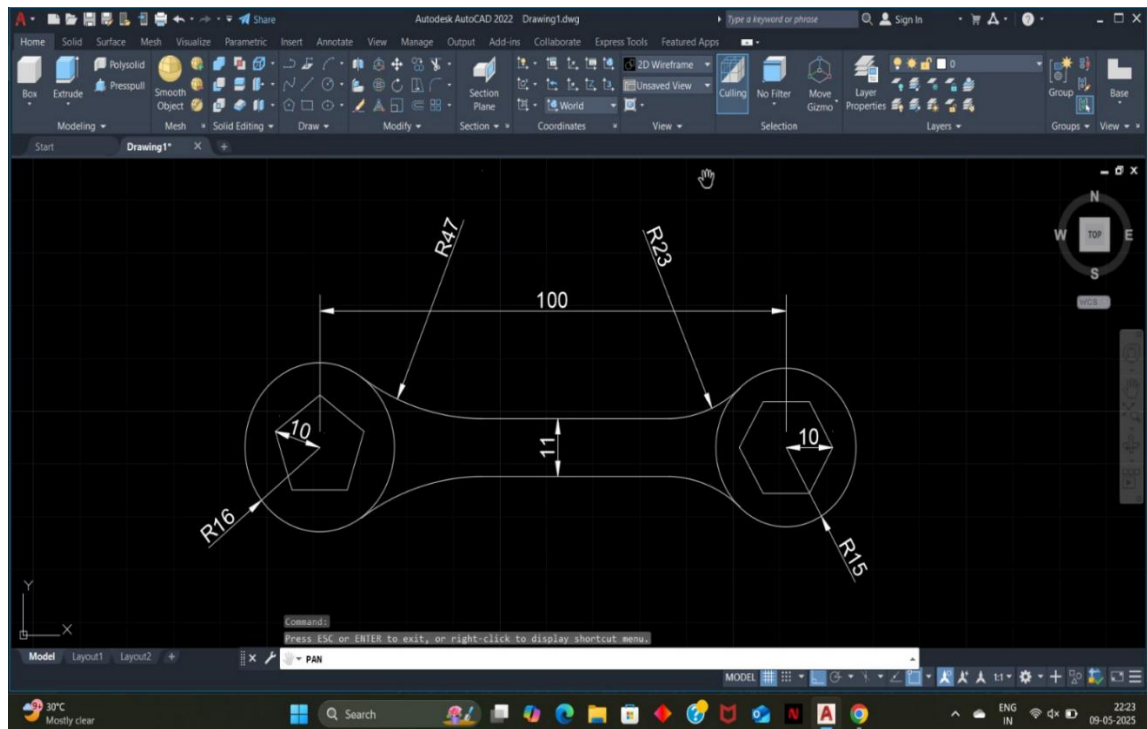
ASSIGNMENT OF SOME MECHANICAL MACHINE COMPONENT USING AUTOCAD 2D

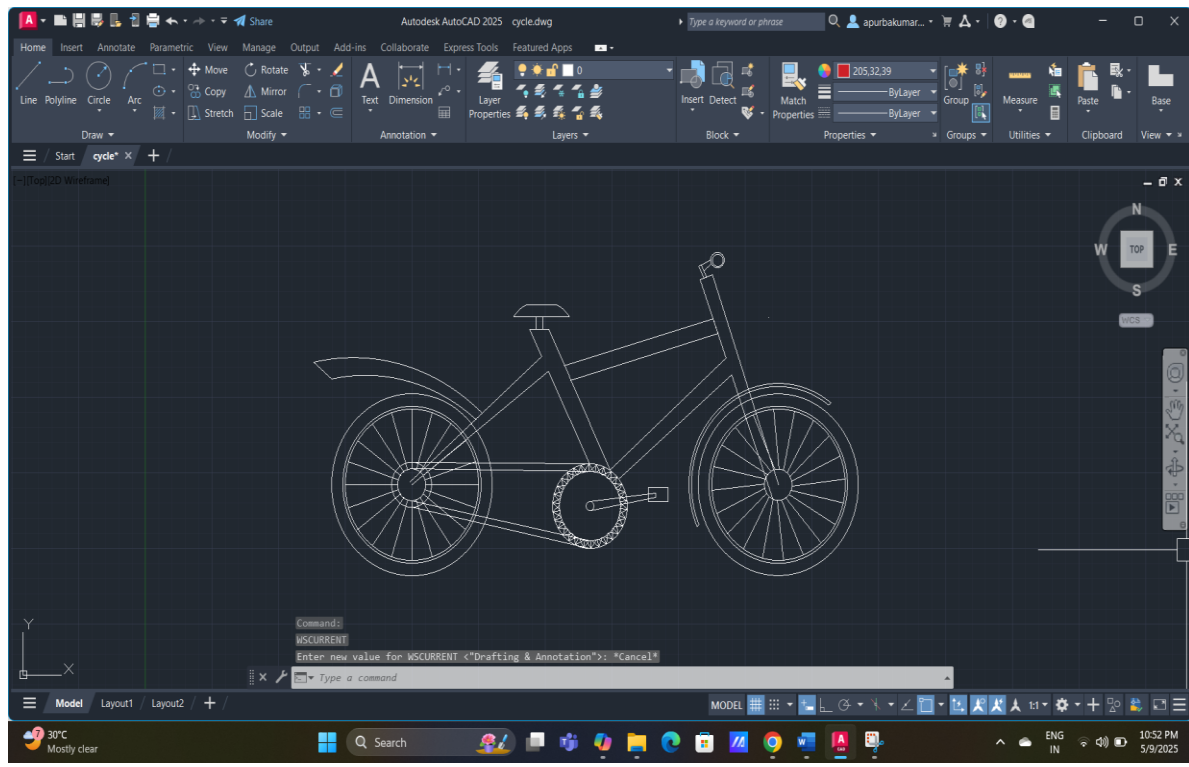






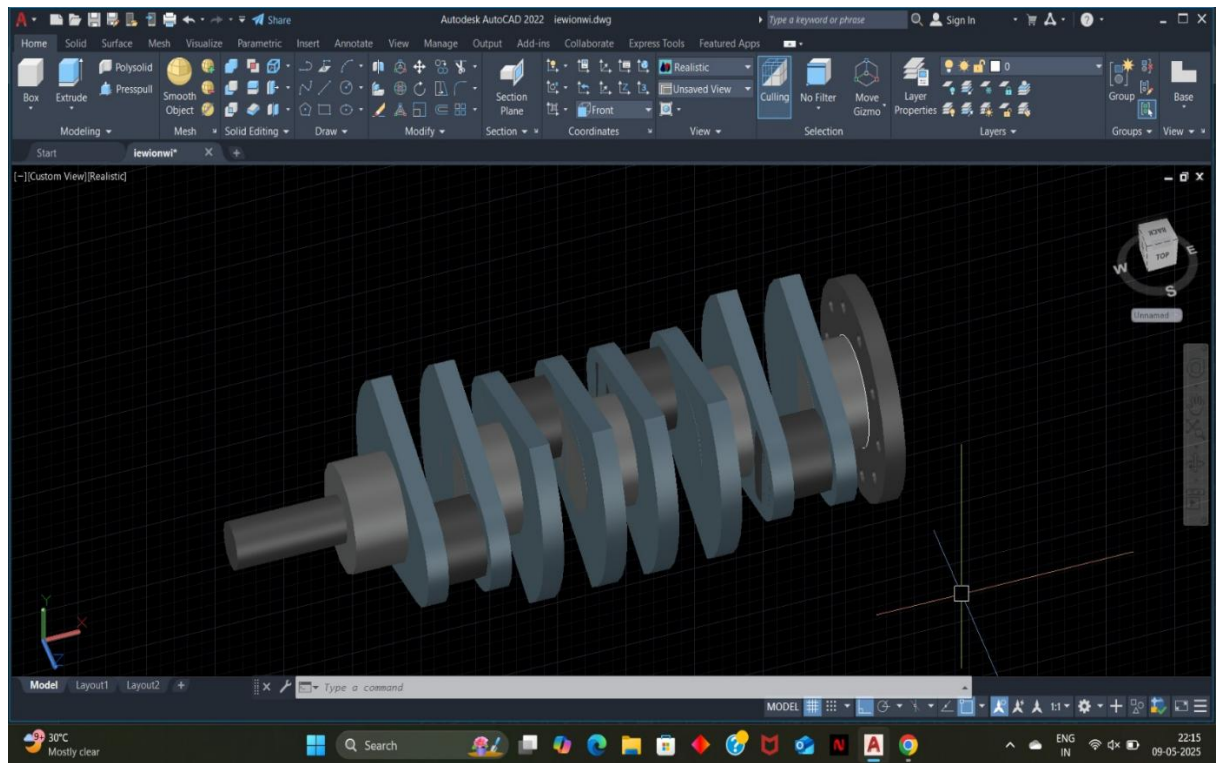




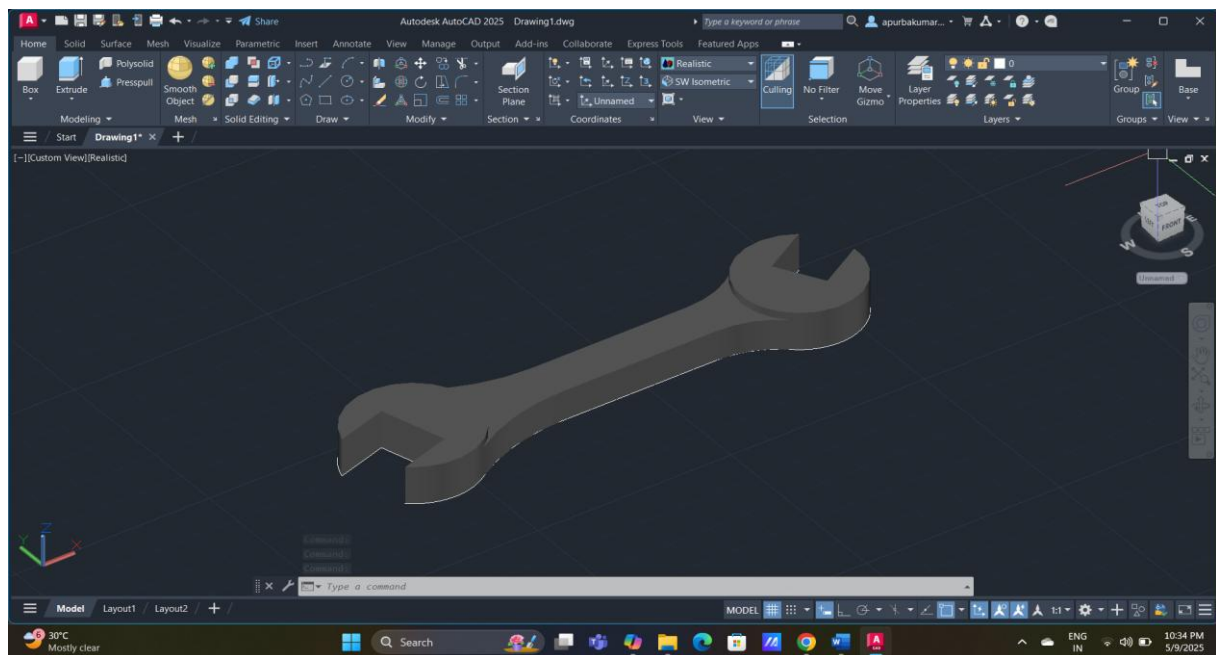


ASSIGNMENT OF SOME MECHANICAL MACHINE COMPONENT USING AUTOCAD 3D

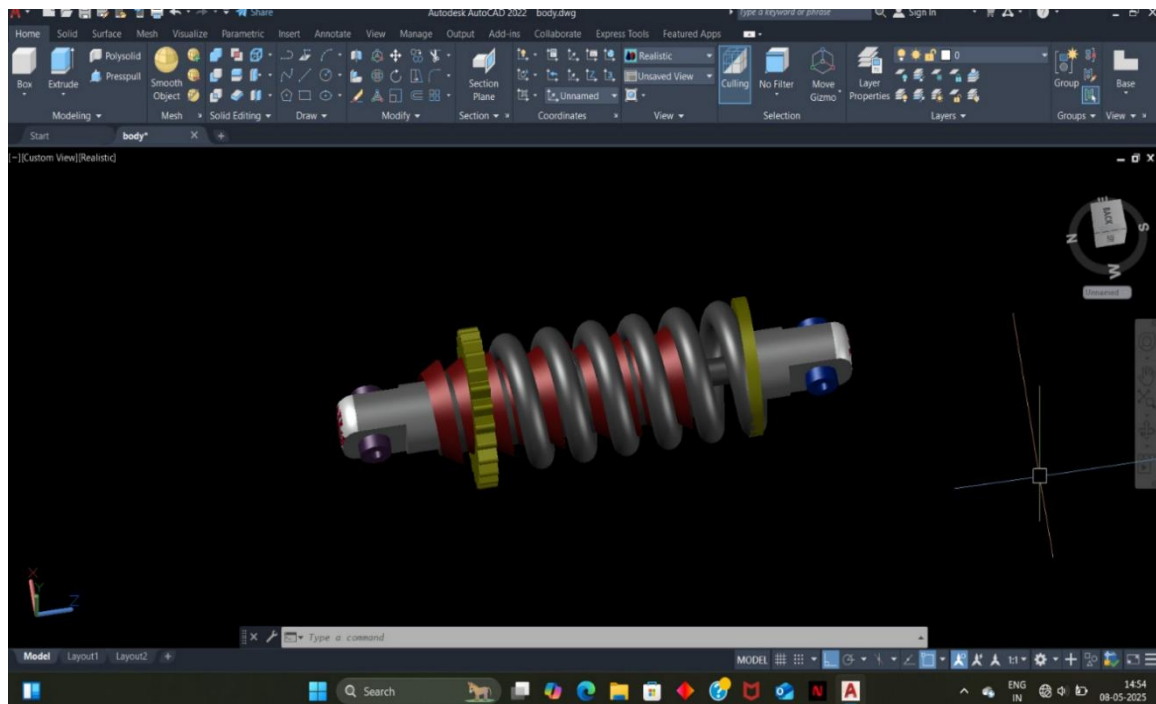
1. 4 CYLINDER PISTON ENGINE :



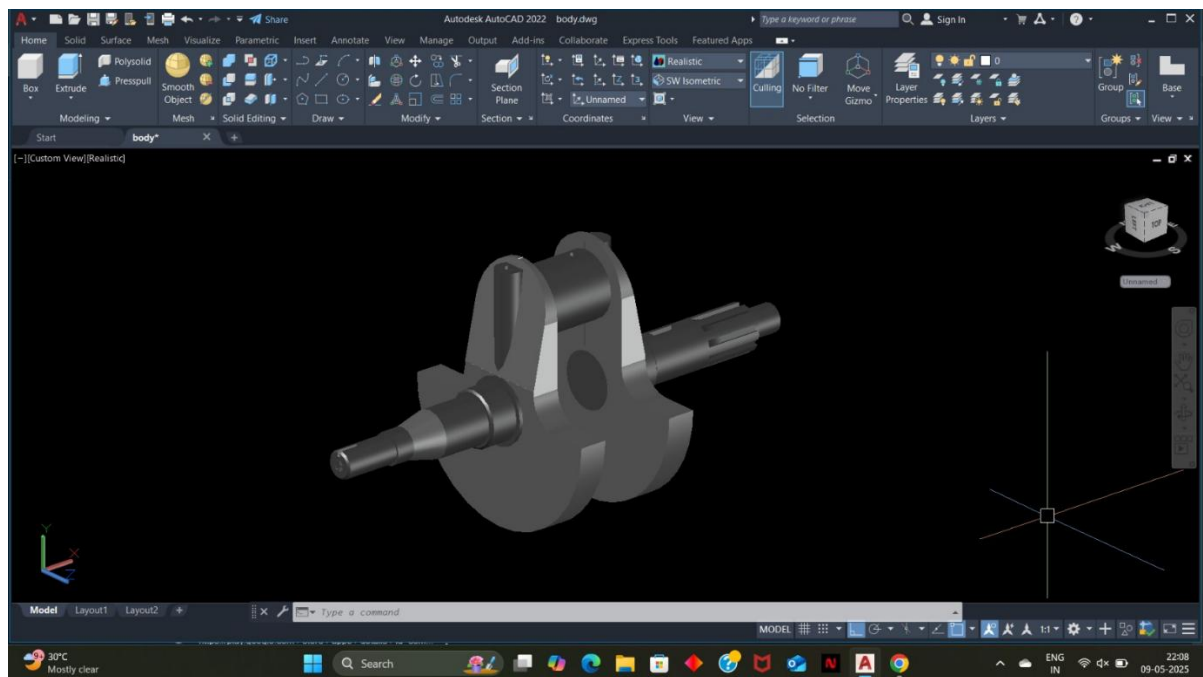
2.WRENCH:-



3. SHOCK ABSORBER



4. CRANK SHAFT



DESCRIPTION OF EACH PART

1. Fuselage

- The main body of the aircraft, housing the cockpit, avionics, fuel tanks, and weapons systems.
- Designed for aerodynamic efficiency to reduce drag and maximize speed.

2. Cockpit

- The pilot's control center, featuring advanced avionics, flight instruments, and heads-up displays (HUD).
- Equipped with ejection seats for emergency escape.

3. Wings

- Provide lift and stability during flight.
- Often designed with advanced aerodynamics to enhance maneuverability.

4. Tail Assembly

- Includes vertical and horizontal stabilizers to maintain stability and control.
- Used for pitch and yaw adjustments.

5. Engine

- Typically a jet turbine or turbofan engine providing thrust for high-speed flight.
- Some advanced fighters have thrust-vectoring engines for improved maneuverability.

6. Landing Gear

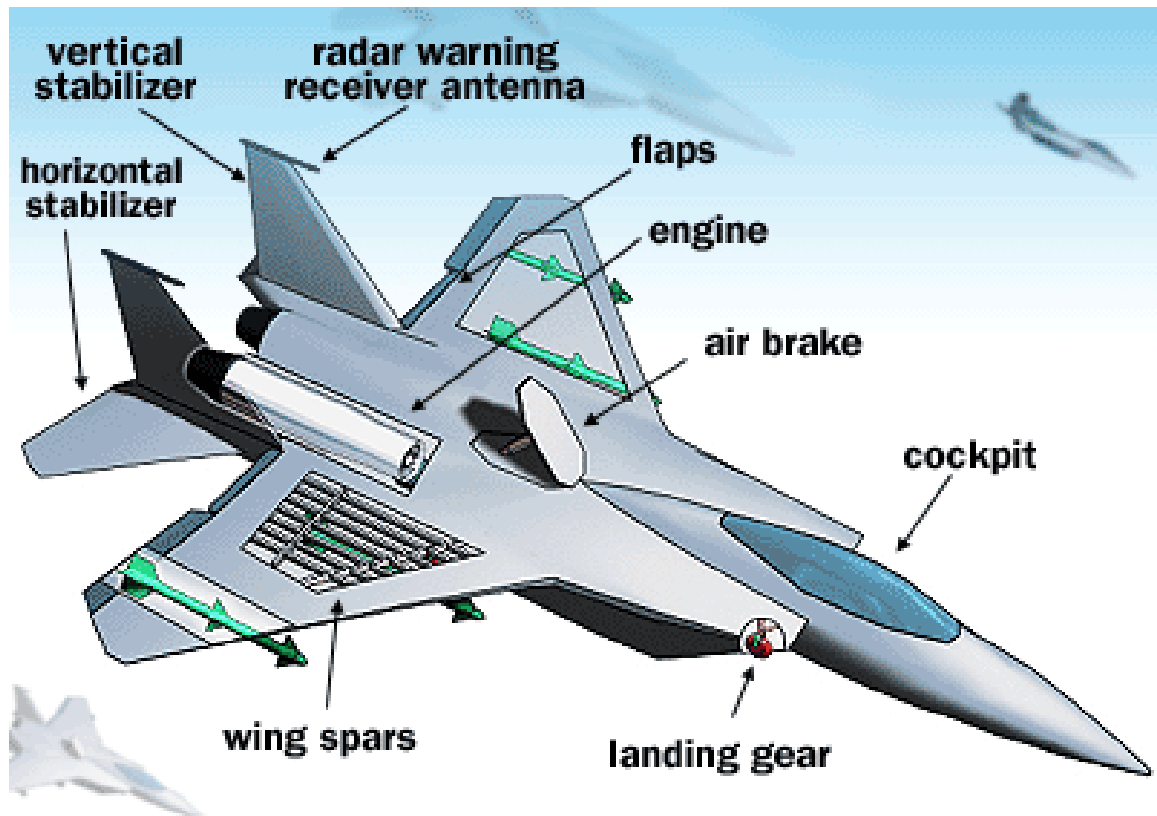
- Retractable wheels allowing the aircraft to take off and land.
- Designed to withstand high-impact landings.

7. Weapons Systems

- Includes guns, missiles, bombs, and other armaments for combat missions.
- Some fighter jets have stealth capabilities to enhance survivability.

8. Radar and Avionics

- High-tech systems used for navigation, targeting, and communication.
- Modern jets feature advanced radar systems for detecting and tracking enemies.

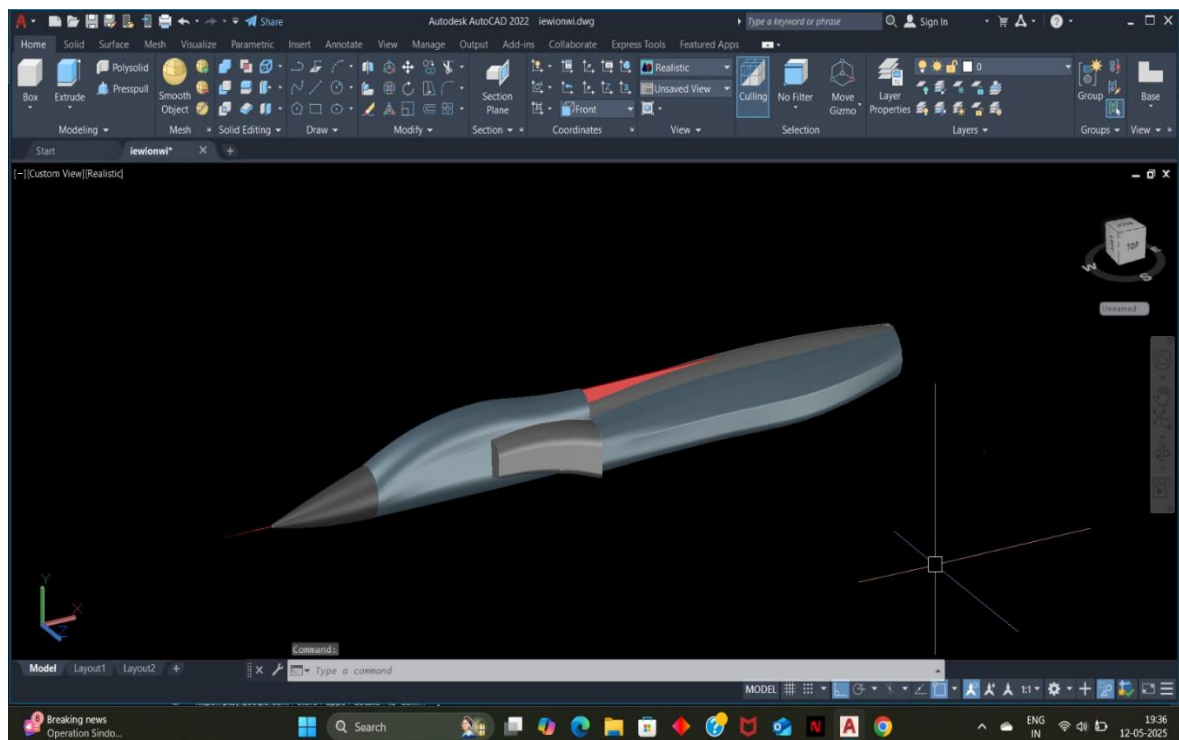


DESIGNING PARTS

- Assumed Dimensions (in mm):
 - Fuselage: 200 mm long, 30 mm max diameter (at the widest point), tapering to 5 mm at the nose and 10 mm at the tail.
 - Wings: 80 mm span (each wing 40 mm from the fuselage centerline), 20 mm chord (width), 5 mm thick.
 - Tail (Horizontal Stabilizer): 40 mm span (20 mm each side), 15 mm chord, 3 mm thick.
 - Vertical Stabilizer: 30 mm height, 15 mm chord, 3 mm thick.

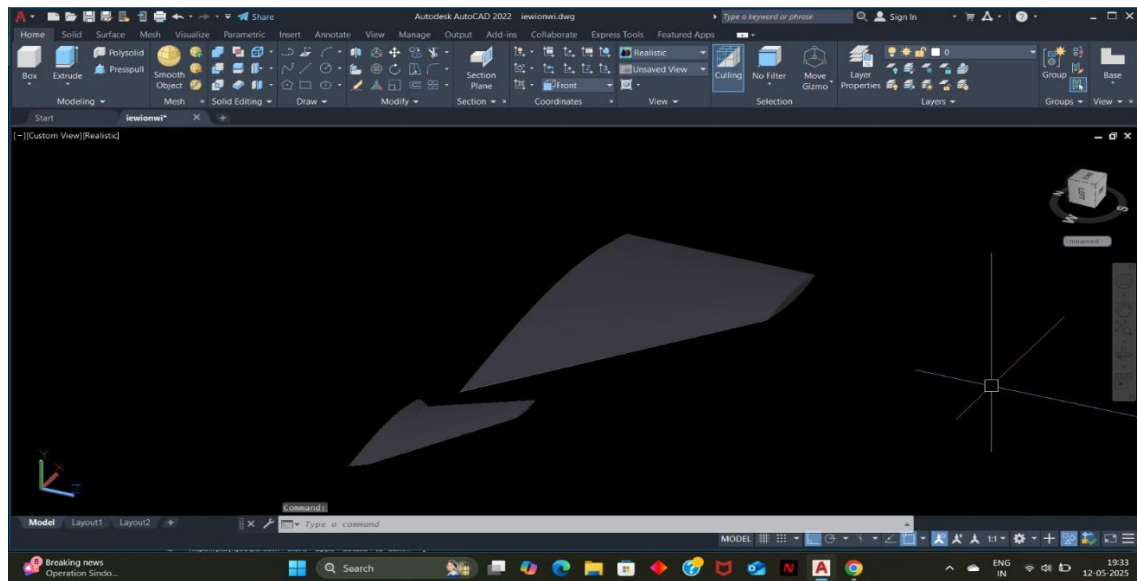
Create the Fuselage (Main Body):

- Draw the side profile of the fuselage using a polyline.
 - Command: POLYLINE
 - Start at (0,0), then enter points: (0,15), (100,15), (180,10), (200,5), (200,0), (180,0), (100,0), (0,0) to close the shape. This creates a tapered profile.
- Revolve the profile to create the fuselage.
 - Command: REVOLVE
 - Select the polyline, press Enter, specify the axis as the X-axis (type 0,0,0 and 1,0,0 for the axis points), and set the angle to 360 degrees.



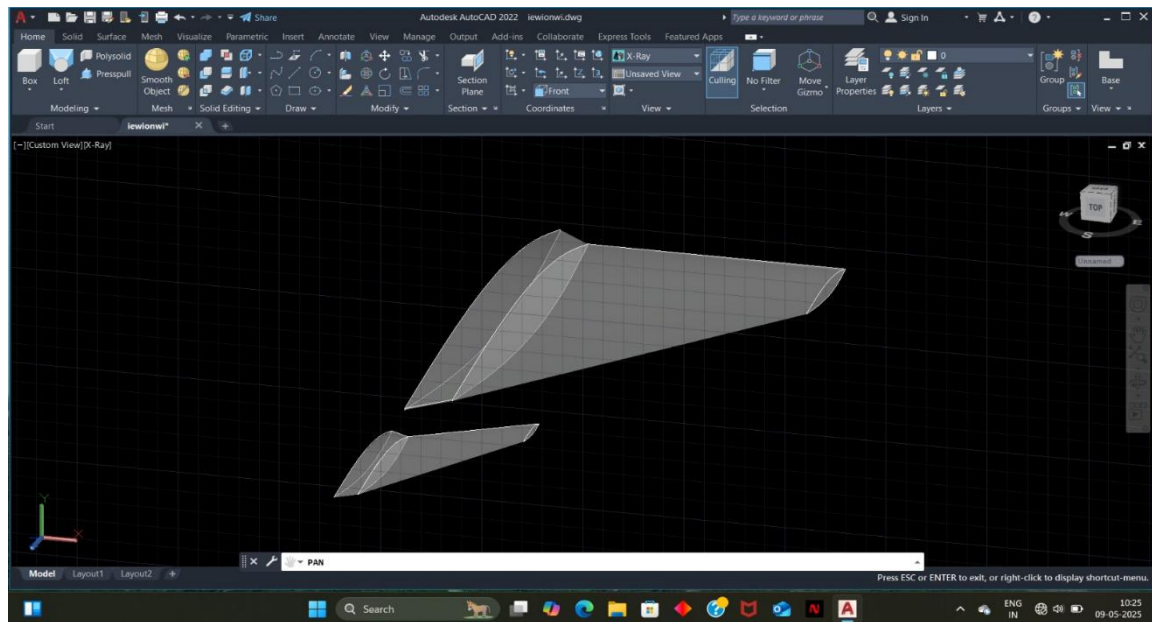
Create the Wings:

- Draw the 2D profile of one wing on the XY plane.
 - Command: POLYLINE
 - Start at (90,0), then: (90,40), (110,40), (110,0), (90,0) to close. This creates a rectangular wing 20 mm wide and 40 mm long.
- Extrude the wing to give it thickness.
 - Command: EXTRUDE
 - Select the polyline, press Enter, specify height as 5 mm.
- Position the wing on the fuselage.
 - Command: MOVE
 - Move the wing to align with the fuselage (e.g., move it to (0,15,0) so it sits on the fuselage's side).
- Mirror the wing to the other side.
 - Command: MIRROR
 - Select the wing, specify the mirror line along the fuselage centerline (e.g., from (0,0,0) to (200,0,0)), and keep the original.



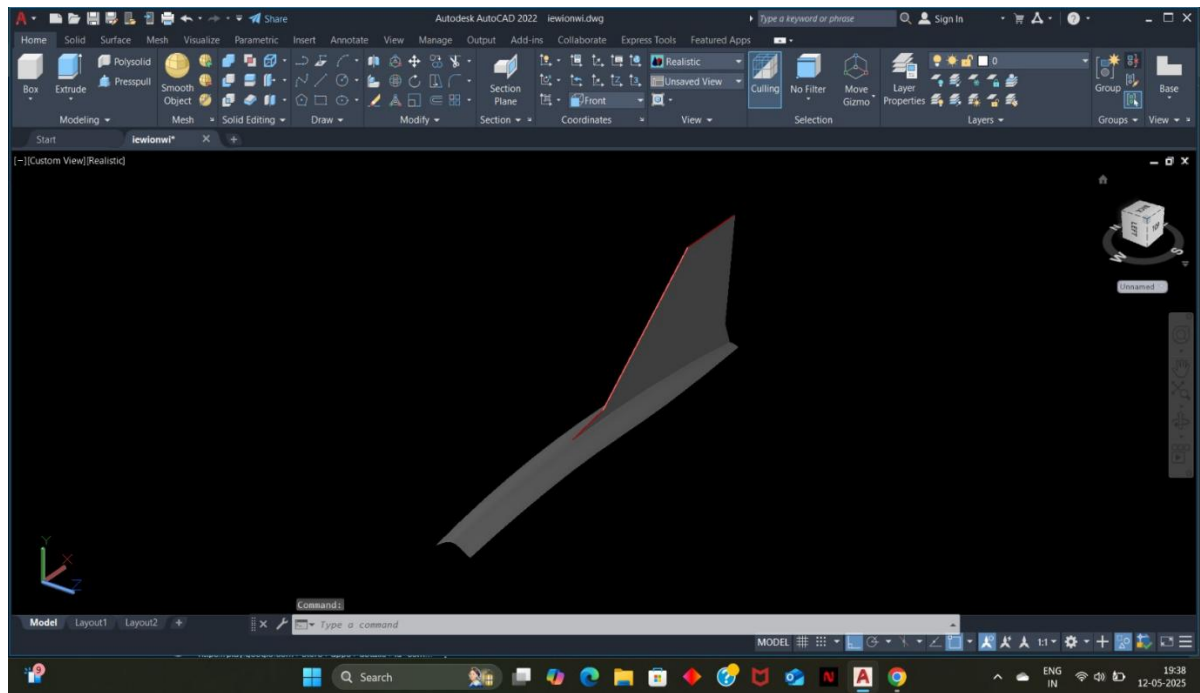
Create the Horizontal Tail (Stabilizer):

- Draw the 2D profile.
 - Command: POLYLINE
 - Start at (170,0), then: (170,20), (185,20), (185,0), (170,0) to close. This creates a 15 mm wide, 20 mm long tail section.
- Extrude the tail.
 - Command: EXTRUDE
 - Select the polyline, specify height as 3 mm.
- Position it at the rear of the fuselage.
 - Command: MOVE
 - Move to (0,15,0) to align with the fuselage top.
- Mirror to the other side.
 - Command: MIRROR
 - Mirror along the fuselage centerline.



Create the Vertical Stabilizer:

- Draw the 2D profile.
 - Command: POLYLINE
 - Start at (170,0), then: (170,30), (185,15), (185,0), (170,0) to close. This creates a triangular stabilizer 30 mm tall and 15 mm wide.
- Extrude it.
 - Command: EXTRUDE
 - Select the polyline, specify height as 3 mm.
- Rotate and position it.
 - Command: ROTATE
 - Rotate 90 degrees around the X-axis to make it vertical.
 - Command: MOVE
 - Move to the rear of the fuselage (e.g., (0,0,15)).



Create the missile :

AutoCAD Commands for the Missile Model:

1. Main Body (Cylindrical Section):

- POLYLINE: Draw a rectangular profile (e.g., 150 mm long, 10 mm half-diameter).
- REVOLVE: Revolve the profile 360° around the X-axis to create a cylinder.

2. Nose Cone:

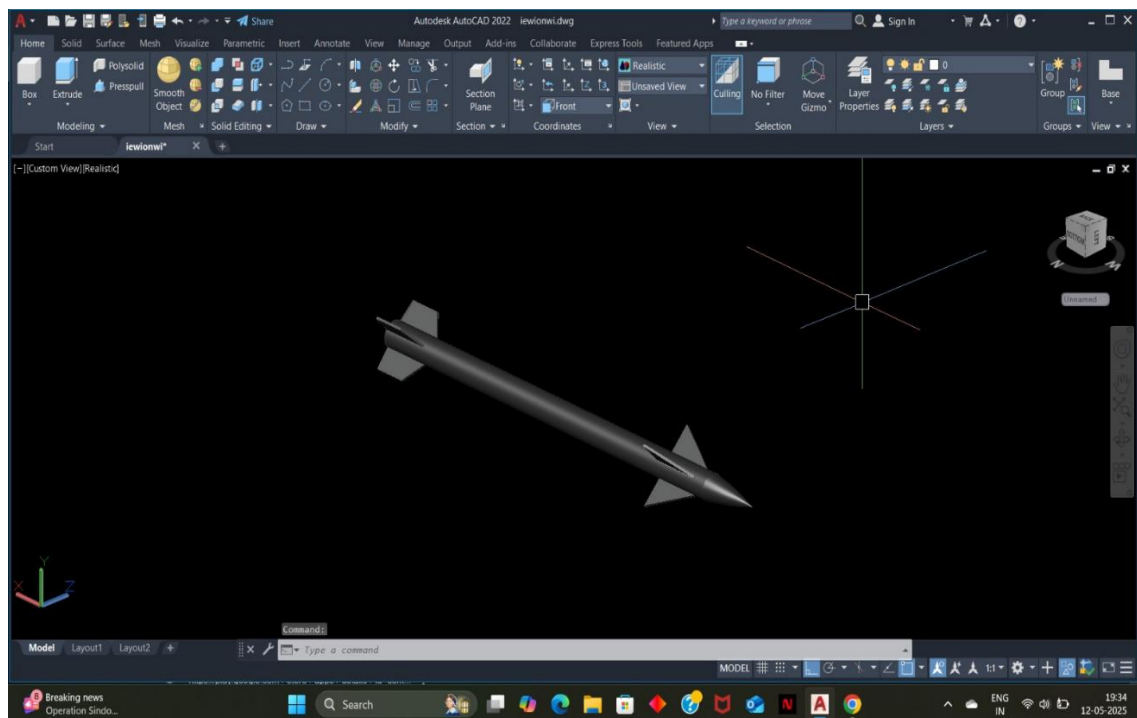
- POLYLINE: Draw a triangular profile for the nose (e.g., 30 mm long, 10 mm half-diameter at the base).
- REVOLVE: Revolve 360° around the X-axis.
- UNION: Combine the nose cone with the body.

3. Fins (4 Total):

- POLYLINE: Draw a rectangular fin profile (e.g., 30 mm long, 20 mm high).
- EXTRUDE: Extrude to 3 mm thickness.
- MOVE: Position the fin on the body.
- ROTATE with COPY: Rotate and copy the fin at 90°, 180°, and 270° to create four fins.

4. Combine Parts:

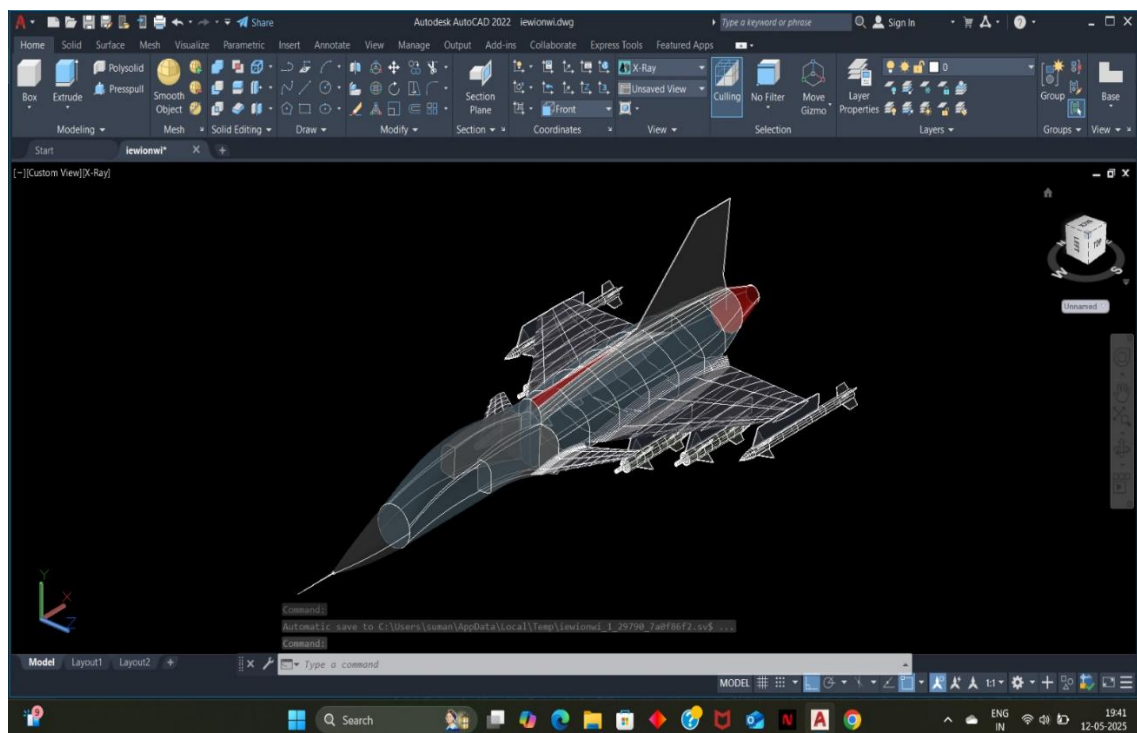
- UNION: Combine the body, nose cone, and fins into a single solid.



➤ Combine All Parts:

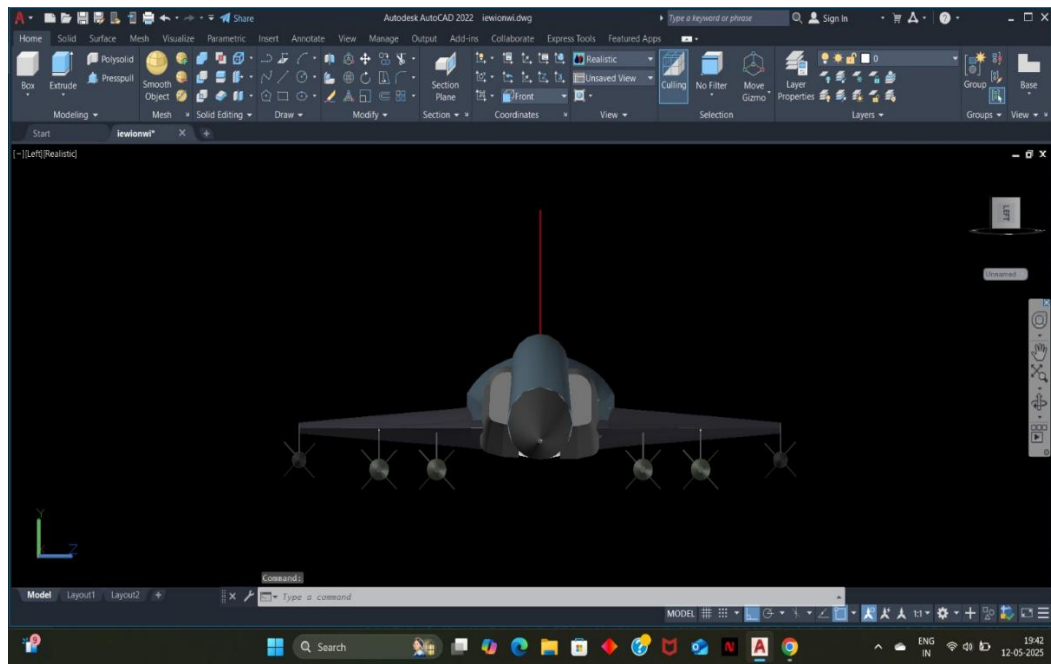
- Use the UNION command to combine all solids.

- Command: UNION
- Select the fuselage, wings, horizontal stabilizer, and vertical stabilizer, then press Enter.
- **Set Visual Style (Optional):**
- The image shows the "Shades of Gray" visual style.
 - Command: VISUALSTYLES
 - Select "Shades of Gray" from the dropdown or type it in.

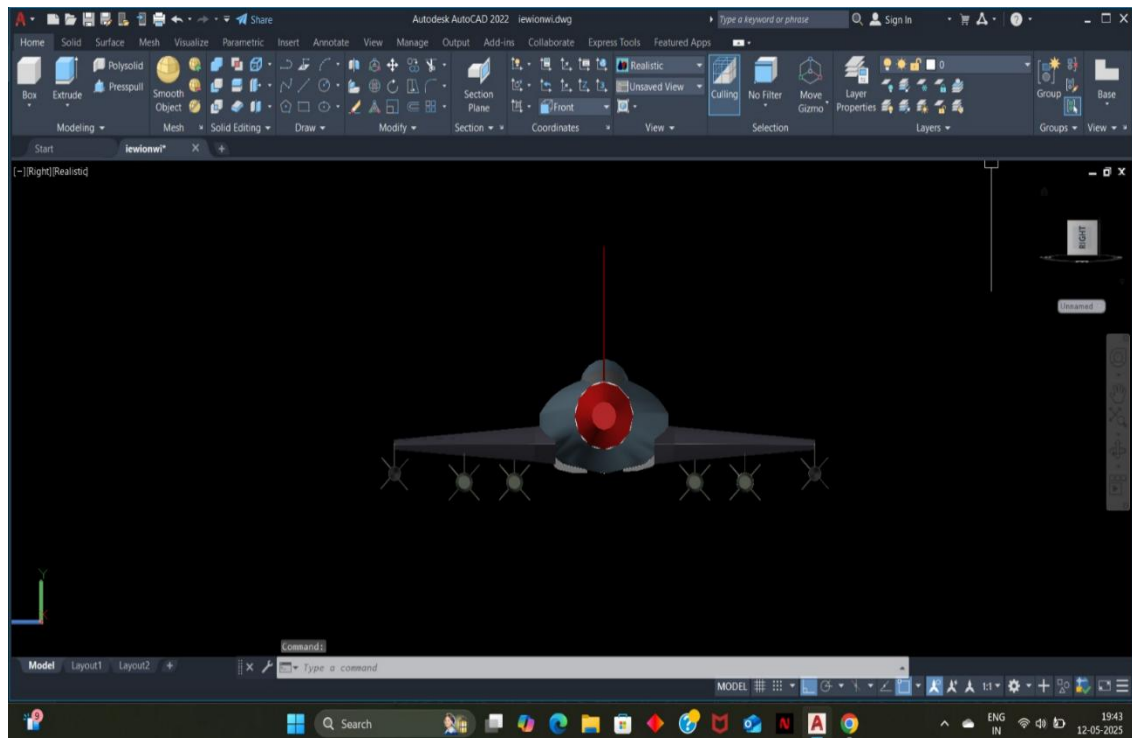


➤ PROJECT PICTURES ALL SIDE VIEW:-

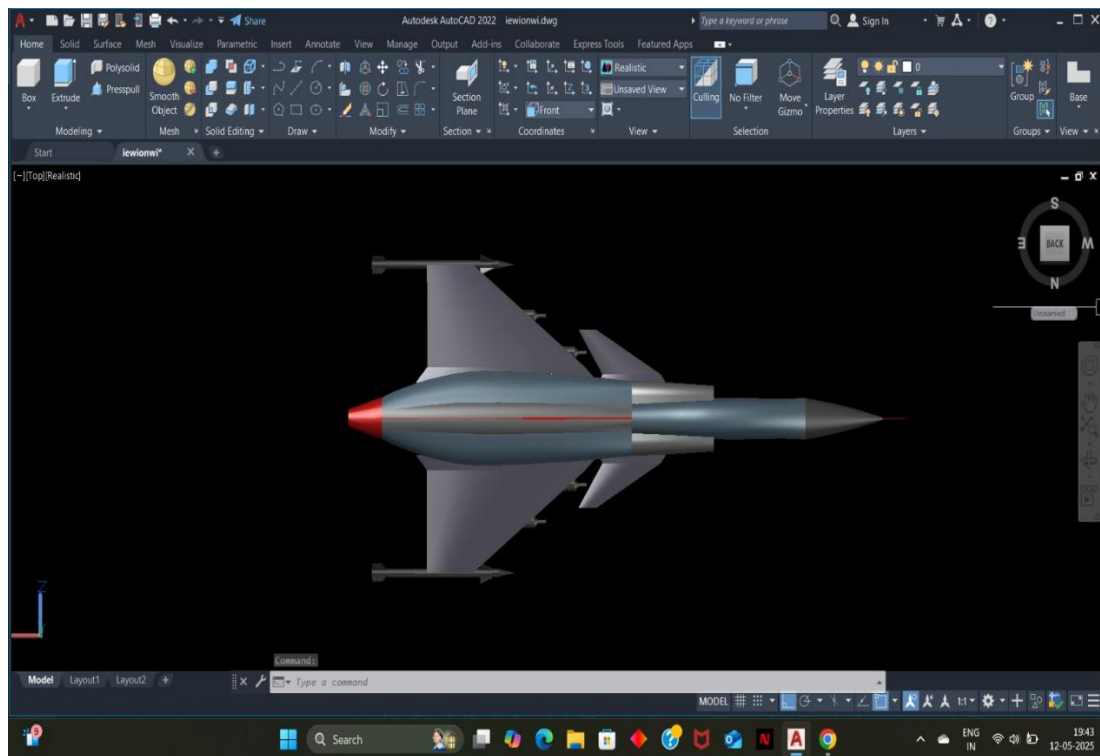
1. FRONT VIEW :



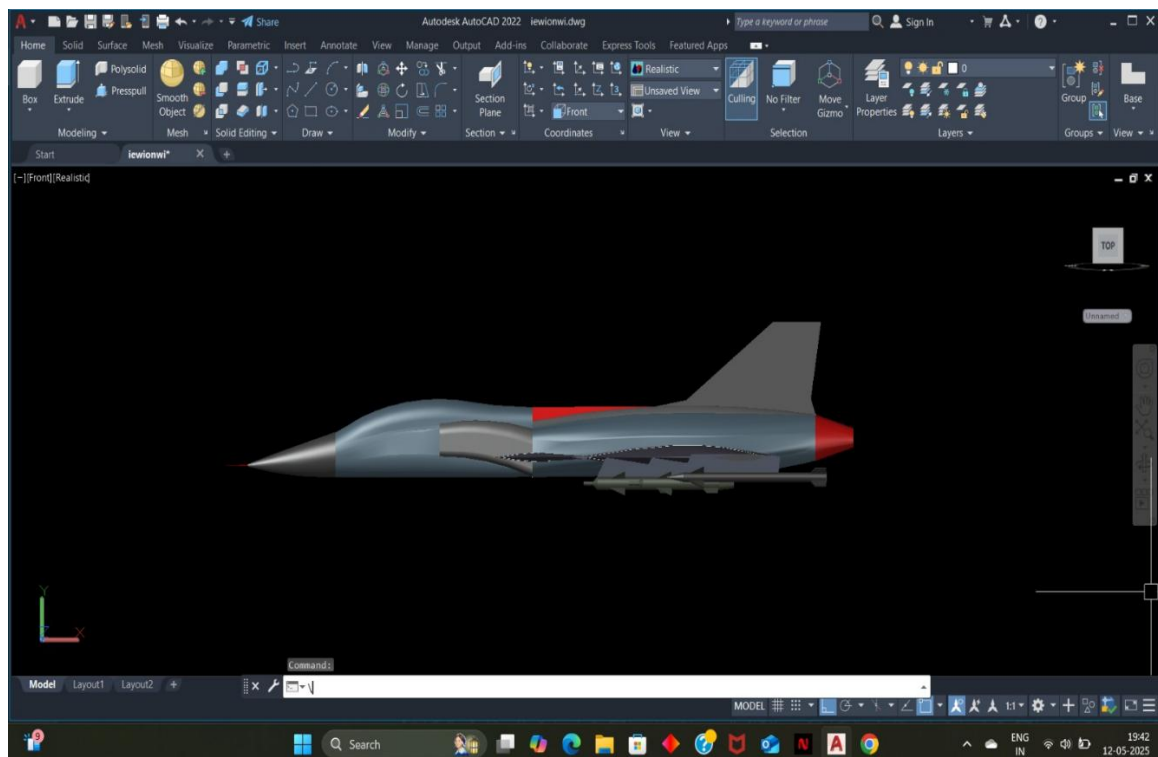
2. BACK SIDE VIEW :-



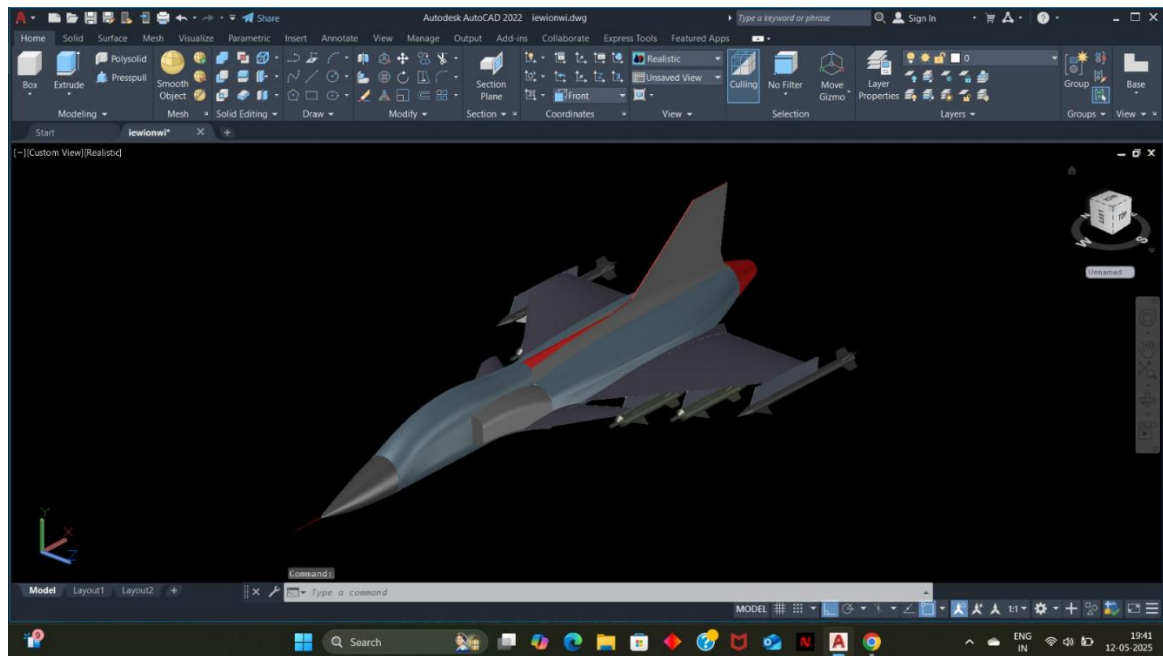
3.TOP SIDE VIEW :-



4.RIGHT SIDE VIEW :-



5. ISOMETRIC VIEW :-



CONCLUSION

❖ AutoCAD 2D and 3D is a leading CAD software for precise technical drawings and detailed models. It boosts productivity, supports diverse industries, and enables collaboration, despite a steep learning curve and cost. Essential for architecture, engineering, and design, it drives innovation and accuracy.

❖ Conclusion of Fighter Jet Plane Design in AutoCAD:

AutoCAD's powerful 2D and 3D tools enable precise design of fighter jet planes, supporting detailed schematics, aerodynamic modeling, and complex assemblies. It ensures accuracy, aids performance analysis, and enhances visualization for aerospace applications. Despite a learning curve and resource demands, AutoCAD streamlines workflows and fosters collaboration, making it essential for innovative, high-precision fighter jet design.

*****PROJECT REFERENCE TAKEN FROM FIGHTER JET- SU 35***

THE END

